

HAT-P-37b

R-2650

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_160725 · created 2026-08-29T23:24:14+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457594.7549 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.049
Transit depth (Rp/Rs)^2	2.71 +/- 1.62 [%]
Orbital Inclination (inc)	89.76 +/- 2.4
Airmass coefficient 1 (a1)	0.12 +/- 0.12
Airmass coefficient 2 (a2)	2.98 +/- 1.56
Scatter in the residuals of the lightcurve fit is	231.4 %
Best Comparison Star	#2 - [286, 288]
Optimal Aperture	4.31
Optimal Annulus	14.69
Transit Duration (day)	0.102 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.049 vs 0.1394 (deviation 0.26σ)
Duration fit vs published	0.102 d vs 0.0983 d (deviation 0.16σ)
Mid-time O–C	-4.8 min
Depth SNR	1.6
Residual scatter	231.4 %

Light curve

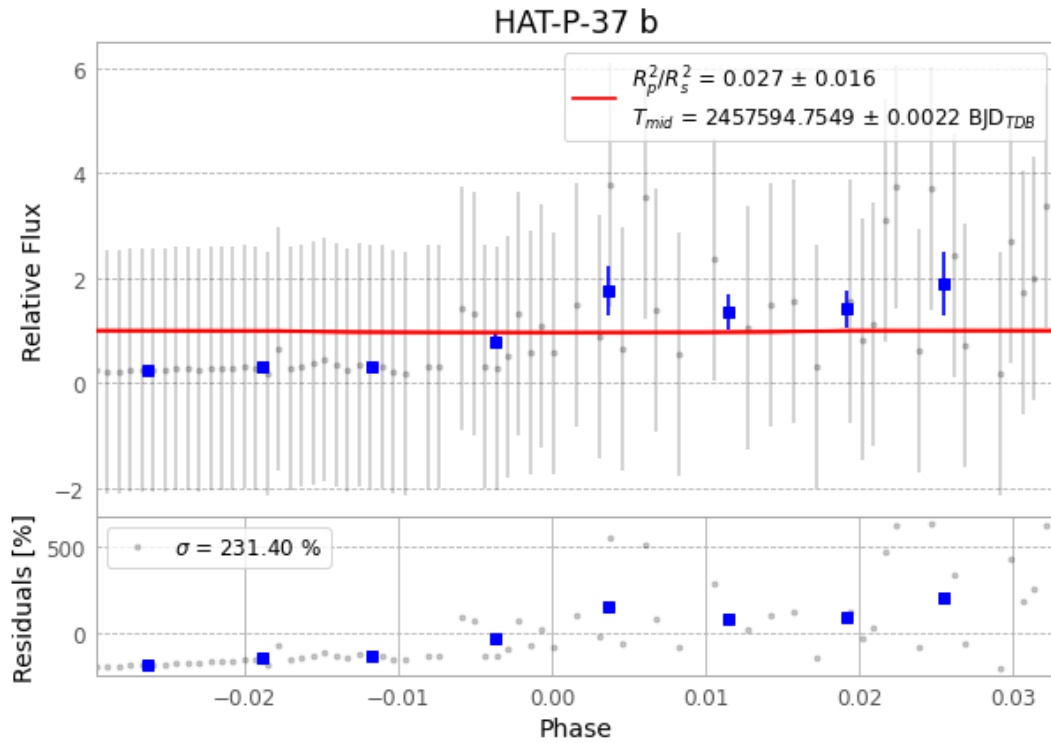


Figure 1 — FinalLightCurve_HAT-P-37 b_2016-07-24.png (SHA-256 c69e94c774f95161...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [127, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5b2cc3e26cfc257e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

85 FITS frames (56 MB), HATP-37160725040312.FITS → HATP-37160725081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-160725.log (e03cc1466e0e...), FinalLightCurve_HAT-P-37 b_2016-07-24.png (c69e94c774f9...), FinalLightCurve_HAT-P-37 b_2016-07-24.csv (5b52512333b4...)

Compute resources

Wall time 236.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:24Z.